



EARTHDATA Data Access Flowchart

Help! If you have any questions, consult the [Earthdata Forum](#)

Start: How can I get specific data from NASA? (example: precipitation over Greece in June 2025)

FIND DATA

Goal: Identify the best data collection for your needs (example: GPM_3IMERGHH, v07)

STEP 1: From earthdata.nasa.gov, enter the name of your desired dataset into the search bar and navigate to that dataset's information page (example)

Decide if it is right for your use by reading its User Guide

Common ways to find a dataset

NASA Earthdata Website
earthdata.nasa.gov

- + Full data catalog & supplemental material
- Information overload

Journal article citations

- + View practical use and DOI stated
- Version + time period not guaranteed

Word of mouth

- + Direct recommendation
- May not apply to your current needs

Search engine

(Google, Bing, etc.)

- + Results show popular datasets
- Best dataset for you may lower on results list

AI agent

(ChatGPT, Claude, Gemini, etc.)

- + Customized request yields customized result
- Result may not be accurate due to algorithm input

DOWNLOAD or STREAM DATA

Goal: Download or stream the variable (example: precipitation)

STEP 2: Prepare your list of data needs

- ☐ dataset name
- ☐ dataset version
- ☐ variable(s)
- ☐ time range/frequency
- ☐ spatial bounds

Do I prefer a web-based or code-based method?

Web-based

Code-based

Why AWS us-west-2 region?

This is the same region in which Earthdata's cloud archive resides. Currently a user making use of compute within the same region can take advantage of direct access to data without egress costs.

OPTION A: Click on the Data Access button and choose Earthdata Search.

Select parameters, such as spatial and temporal bounds (example: polygon around Greece; Jun 1-30, 2025).

Click on the "Download All" button and choose how you download your data

The [Earthdata Download](#) application is necessary for downloading a large number of files from Earthdata Search

Download

OPTION B: Subset and download with OPeNDAP



Tutorial 1

Download

View more options for data browsing, exploration, access and other tasks in this chart, [Which Data Tool Is Right For You?](#)

"Download" means the files are saved to your computer
"Stream" means the file bytes are accessed over the internet and read directly into computer memory, and NOT saved to your computer

OPTION C: Use S3 direct access to stream data within the cloud



Tutorial 1 | 2 | 3

Stream

OPTION D: Download subsetted data with harmony-py



Tutorial 1 | 2

Download

OPTION E: Download all data with earthaccess



Tutorial 1 | 2 | 3

Download

OPTION F: Stream all data with earthaccess



Tutorial 1

Stream

OPTION G: Stream subsetted data with OPeNDAP



Tutorial 1 | 2

Stream

Is my computing in the AWS cloud (us-west-2 region)?

Yes

No

"All" refers to the full file
"Subset" refers to a portion of the file

Am I using the Python programming language?

Yes

No

OPTION H: Download subsetted data by Harmony API with curl



Tutorial 1

Download

OPTION I: Download data using the command line with curl or wget



Tutorial 1 | 2

Download

End: I get the data I need 😊

Recommendations based on task



I need to complete a project soon, I need to get data one time and store it on my computer to analyze

Recommendation: Choose OPTION A or B

Tip: Confirm your amount of storage space is enough for the data



I am part of an operational team that requires a near-constant stream of data to integrate into our products

Recommendation: Choose OPTION G

Tip: Set up a script to automate regular data access



I am on a team that analyzes a region of interest every few months, I need to get the same data intermittently and can store it on local resources

Recommendation: Choose OPTION D or H

Tip: Instead of repeatedly using the web interface, create a script that subsets and downloads the data

I am a user of Python. Which libraries are best for accessing Earthdata?

- [Harmony.py](#) and [earthaccess](#) are recommended

Recommendation: Choose OPTION D, E, or F



I am *not* a Python user. Which programmatic method is best for accessing Earthdata?

- Harmony API and OPeNDAP are recommended

Recommendation: Choose OPTION H or I



More web-based tools available from earthdata.nasa.gov

WEB-BASED TOOL	BEST FOR	DISCIPLINE / TOPIC
Earthdata Search	Exploring, searching and downloading satellite data	All Earthdata holdings
Worldview	Exploring & visualizing satellite data	Time-critical application areas such as wildfire management, air quality measurements, flood monitoring
Earthdata GIS	Visualizing select satellite data	Featured collections include: TEMPO, SWOT, PACE, OPERA RTC-S1, GPM IMERG, NASA Disasters, NASA Landslides, POWER, TROPOMI, Carbon Monitoring
FIRMS Fire Information and Resource Management System	Visualizing near-real time fire data	Fire science, active fires
Vertex	Finding and pre-processing synthetic aperture radar data	Synthetic aperture radar (SAR) data
Giovanni Geospatial Interactive Online Visualization AND aNalysis Infrastructure	Plotting time and area averages of atmospheric, hydrologic, cryospheric and oceanic data	Select datasets covering atmospheric chemistry & dynamics, aerosols, cryosphere, hydrology, ocean biology, oceanography, water & energy cycle
AppEEARS Application for Extracting and Exploring Analysis-Ready Samples	Reducing data volume and preparing analysis-ready land process related data for download	Popular datasets from ECOSTRESS, EMIT, HLS, SRTM, MODIS, VIIRS, SMAP, Daymet, GWP
OpenAltimetry	Visualizing & downloading surface elevation data	ICESat & ICESat-2 observations of land ice height, sea ice height, land and vegetation height, sea ice freeboard, ocean surface height, inland water surface data
Field Campaign Explorer	Exploring data from multiple sub-orbital instruments	Atmospheric observations from HS3, CPEX-AW, OLYMPEx, IMPACTS, GOES-R-PLT
SOOT Sub-Orbital Order Tool	Finding and downloading data from sub-orbital instruments	Atmospheric observations from field campaigns that include radiation budget, clouds, aerosols, and tropospheric composition